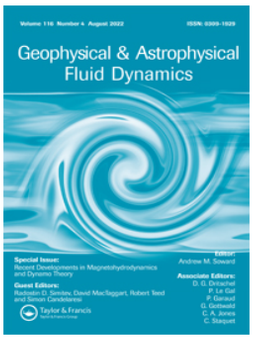

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Rigorous results on conserved and dissipated quantities in ideal MHD turbulence

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ABSTRACT
We review recent mathematical results on the theory of ideal MHD turbulence. On the one hand, we explain a mathematical version of Taylor's conjecture on magnetic helicity conservation, both for simply and multiply connected domains. On the other hand, we describe how to prove the existence of weak solutions conserving magnetic helicity but dissipating cross helicity and energy in 3D Ideal MHD. Such solutions are bounded. In fact, we show that as soon as we are below the critical L^3 integrability for magnetic helicity conservation, there are weak solutions which do not preserve even magnetic helicity. These mathematical theorems rely on understanding the mathematical relaxation of MHD which is used as a model of the macroscopic behaviour of solutions of various nonlinear partial differential equations. Thus, on the one hand, we present results on the existence of weak solutions consistent with what is expected from experiments and numerical simulations, on the other hand, we show that below certain thresholds, there exist pathological solutions which should be excluded from physical grounds. It is still an outstanding open problem to find suitable admissibility conditions that are flexible enough to allow the existence of weak solutions but rigid enough to rule out physically unrealistic behaviour.

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1. Introduction

This contribution reviews recent mathematical developments on Onsager's theory of turbulence in the context of MHD. We discuss the use of singular (weak) solutions in the description of MHD turbulence at high magnetic Reynolds numbers. Special emphasis is put on three classical conserved quantities of ideal MHD: total energy, cross helicity and magnetic helicity. We begin by discussing the motivation behind the use of weak solutions in MHD.

In numerous experimental and numerical setups, both astrophysical and laboratory plasmas tend to undergo a relaxation towards a small number of preferred states, largely independently of initial conditions (see, e.g. Ortolani and Schnack 1993, Yeates 2020). An

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archetypical example consists of linear force-free states $\nabla \times \boldsymbol{B} = \alpha \boldsymbol{B}$. A variational explanation for the linear force-free state was proposed by Chandrasekhar and Woltjer (1958).

Woltjer (1958) suggested a different variational problem where magnetic energy is minimised under the constraint that the magnetic helicity $\int_\Omega \mathbf{A}(x,t) \cdot \boldsymbol{B}(x,t)\,\mathrm{d}x$ is fixed. He computed formally that the minimisers are, indeed, force-free (see Laurence and Avellaneda 1991 and Komendarczyk 2022 for rigorous studies). Woltjer also showed that in ideal MHD, the magnetic helicity $\int_\Omega \boldsymbol{A}(x,t) \cdot \boldsymbol{B}(x,t)\,\mathrm{d}x$ (with a suitable magnetic potential $\boldsymbol{A}$) is conserved in time whenever Ω is a bounded, simply connected, magnetically closed domain. In such domains, magnetic helicity is in fact gauge-invariant, but in magnetically open volumes, where the magnetic field lines are allowed to thread the boundary, the relative magnetic helicity of Berger and Field (1984), Jensen and Chu (1984) and Finn and Antonsen (1985) is used instead. In *multiply connected* magnetically closed domains, a natural gauge-invariant definition was given by MacTaggart and Valli (2019). We will come back to these issues in Section 3.1.

Moffatt (1969) observed that magnetic helicity measures the linkage of magnetic field lines (see also, e.g. Arnold and Khesin 1998, Berger 1999 and Prior and MacTaggart 2020 for further topological interpretations of magnetic helicity and its variants). Moffatt noted, furthermore, that magnetic helicity is conserved for all volumes $\Omega_i(t)$ moving with the fluid that are magnetically closed at the initial time (see Yeates 2020, pp. 122–123). Moffatt's discovery leads to potentially numerous conserved quantities (subhelicities) $\int_{\Omega_i(t)} \boldsymbol{A} \cdot \boldsymbol{B}\,\mathrm{d}x$ which act as topological constraints of the evolution. This heavy dependence on the initial state is, however, at odds with the observed relaxation of turbulent plasmas (see, e.g. Ortolani and Schnack 1993).

A solution to the dilemma described above was suggested by Taylor (1974). In the presence of even slight resistivity, subhelicity conservation breaks down as a result of magnetic reconnection. Taylor conjectured that magnetic helicity in the whole domain would nevertheless be approximately preserved. Taylor's conjecture is at the heart of Woltjer–Taylor relaxation theory which, again, proposes to minimise magnetic energy while keeping magnetic helicity fixed (see Ortolani and Schnack 1993, Yeates 2020). Woltjer–Taylor relaxation *per se* does not concern us in this paper, but we will discuss Taylor's conjecture itself. Berger (1984) has shown that under physically realistic assumptions, in dynamical timescales, magnetic energy dissipates very much faster than magnetic helicity. As one of the main themes of this work, we discuss a natural mathematically precise version of Taylor's conjecture by Faraco *et al.* (2022), see Theorem 3.1.

At the heart of Moffatt's theorem and Taylor's recourse to positive resistivity in magnetic relaxation theory is Alfvén's theorem (Alfvén 1942). Alfvén noted that in ideal MHD, the magnetic field is frozen into the fluid, thus prohibiting changes of topology. As a related result, the magnetic flux through a surface moving with the fluid is conserved (in particular, magnetically closed Lagrangian fluid volumes remain magnetically closed in time). Both of these results are commonly referred to as Alfvén's theorem. The theorem is often assumed to hold approximately at small nonidealities (see, e.g. the discussion on p. 1 of Eyink 2011), but to our understanding, this seems difficult to reconcile with the observed rates of magnetic reconnection in various turbulent systems, in particular in solar flares (Eyink 2011, Eyink *et al.* 2011, Jafari and Vishniac 2019, Lazarian *et al.* 2020). The physical breakdown of Alfvén's theorem in high *Rm* turbulence was predicted by Eyink and Aluie (2006).

In the seminal work Eyink *et al.* (2013), the breakdown of magnetic flux conservation was observed in a simulation. The apparent contradiction with Alfvén's theorem can be explained by the fact that *the textbook derivation of Alfvén's theorem tacitly assumes that the fluid velocity and the magnetic field be laminar (differentiable)*. In Eyink *et al.* (2013), due to the roughness of the velocity field, the classical notion of Lagrangian particle trajectory ceases to be applicable; there are, then, no canonical fluid element trajectories that the magnetic field could freeze into. The velocity field undergoes *spontaneous stochasticity* (Bernard *et al.* 1998), and a stochastic variant of flux-freezing holds (see also Eyink 2011, Eyink *et al.* 2011). In fact, it appears that magnetic reconnection can occur even in regimes where ideal MHD is valid, *as long as the fields are not tacitly assumed to be laminar* (see Eyink *et al.* 2011, Eyink 2015).

A more classical case of breakdown of conservation laws is given by anomalous dissipation. In hydrodynamics, anomalous dissipation refers to the empirically and numerically verified fact that at very high Reynolds numbers $Re \gg 1$, the rate of kinetic energy decay is positive and essentially independent of Re. Anomalous dissipation plays a key role already in Komogorov's K41 theory (see, e.g. Frisch 1995). It has also been observed numerically in MHD (Mininni and Pouquet 2009, Dallas and Alexakis 2014, Linkmann *et al.* 2015).

Onsager (1949) gave a fascinating account of anomalous dissipation in hydrodynamics. He predicted that turbulent flows could dissipate kinetic energy even without the assistance of viscosity, *as long as the solutions are interpreted in a weak (distributional) sense*. Again, it is crucial to observe that the textbook derivation of kinetic energy conservation tacitly assumes the fluid velocity to be laminar (differentiable). For reviews on Onsager's turbulence theory geared towards physicists (see Eyink 2008, 2018, Dubrulle 2019). Onsager conjectured that for C^α Hölder continuous solutions of Euler equations, kinetic energy is always conserved when $\alpha > 1/3$ and in general dissipated when $\alpha < 1/3$. Onsager's conjecture has recently come under tremendous mathematical activity. The conservation part of Onsager's conjecture was proved by Constantin *et al.* (1994) after a partial result by Eyink (1994). The dissipation part was recently proved by Isett (2018) and Buckmaster *et al.* (2019) after a lot of work by the mathematics community, as we briefly discuss in section 4.2. For recent empirical work on Onsager's turbulence theory see Kuzzay (2016).

As explained above, weak solutions are not a mere mathematical artefact; if one is to use ideal MHD equations as a model of turbulence at high magnetic Reynolds numbers, anomalous dissipation of energy and fast magnetic reconnection arguably compel one to use them. Furthermore, it is perfectly conceivable that weak solutions are sometimes all that we have, even if the initial state of the system is laminar – the question about global-in-time smooth solutions is wide open in both ideal and resistive MHD. Nevertheless, a weak Leray – Hopf solution of the Cauchy problem (aptly called a "turbulent solution" by Leray himself) exists whenever the initial datum is square integrable (see section 2.2).

Anomalous energy dissipation, Taylor's conjecture and fast magnetic reconnection motivate a critical study of classical conservation laws for weak solutions at the limit of ideal MHD. This would hopefully bring a rigorous counterpart to the available numerical evidence on these problems, an aim that has guided the authors' work on MHD. Below, we briefly describe some of the concrete mathematical problems that arise in this fashion and the state of the art on them.

A natural rigorous formulation of Taylor's conjecture says that magnetic helicity is conserved at the ideal limit (Caflisch *et al.* 1997, p. 444); see Definition 2.2. This conjecture was

proved by Faraco and Lindberg (2020) in simply connected, magnetically closed domains and by Faraco *et al.* (2022) in multiply connected domains, see Theorem 3.1. The evolution of relative magnetic helicity at the ideal limit is still an open problem in magnetically open domains; numerical evidence for a suitable adaptation of Taylor's conjecture was given by Pariat *et al.* (2015).

In order to account for anomalous dissipation, one would like to construct energy dissipative weak solutions of ideal MHD that arise at the ideal limit. This seems, however, to be beyond currently available mathematical techniques. As a more tractable problem, it is natural to try and construct dissipative solutions of ideal MHD directly at regularity regimes where magnetic helicity is conserved (without considering a regularisation such as the ideal limit). By a result of Kang and Lee (2007) and Aluie (2009), magnetic helicity is conserved by weak solutions whenever they are L^3 integrable (third-order moments are finite).

Faraco *et al.* (2021a, 2021b) have found energy dissipative solutions that are not just L^3, but L^∞ integrable, see Theorem 3.3. Thus, the selective decay of energy and conservation of magnetic helicity is, indeed, observed at the level of weak solutions of ideal MHD. Some of the mathematical techniques of the proof are described in section 4.2.

Another natural mathematical problem is whether the L^3 integrability criterion of Aluie and Eyink and Kang and Lee is optimal. As a first result in this direction, Beekie *et al.* (2020) constructed solutions just above L^2 integrability which fail to conserve magnetic helicity. Buckmaster and Vicol (2021) conjectured that L^3 is, indeed, the threshold for magnetic helicity conservation. The conjecture was recently proved by Faraco *et al.* (2021b), see Theorem 3.2. In view of the mathematical version of Taylor's conjecture and the theorems of Beekie, Buckmaster and Vicol and Faraco, Lindberg and Székelyhidi, at the ideal limit (as compared to general weak solutions of ideal MHD), extra conserved quantities appear, in remarkable agreement with experiments and simulations. As such, the results lend credence to ideal limit as a good admissibility criterion for ruling out physically unrealistic solutions (e.g. ones whose total energy increases in time even without external forcing). We discuss this issue further at the end of section 3.2.

We close the introduction by outlining the structure of the paper. The relevant mathematical definitions are recalled in section 2, and the mathematical results mentioned above are made precise in section 3. The main mathematical tools are Tartar's compensated compactness theory and Nash's method of convex integration; both are discussed in section 4 in the context of fluid dynamics. As another very important facet of compensated compactness theory, Tartar formulated a model for macroscopic quantities in fluid mechanics (a substitute for ensemble averages and other kinds of averaging), see section 4.3. Several remaining open problems are listed in section 5.

2. Weak solutions of ideal and resistive MHD

In this chapter, we recall the relevant definitions on weak solutions of ideal and resistive MHD. The basic idea is that the equations are not assumed to hold as normal partial differential equations but only in an integral sense, tested against suitable test functions; this leaves room for phenomena such as shock waves. We work in smooth, bounded domains $\Omega \subset \mathbb{R}^3$ for simplicity. We begin by quickly recalling the definitions of relevant function

spaces in section 2.1; thorough discussions are found, e.g. in books by Evans (2010) and Temam (1979).

2.1. Function spaces

The most basic function spaces we need are the L^p *spaces* of F. Riesz. When $1 \le p \le \infty$, a (Lebesgue measurable) function $f\colon \Omega \to \mathbb{R}$ belongs to $L^p(\Omega)$ if

$$\|f\|_p \equiv \begin{cases} \left(\int_\Omega |f(\boldsymbol{x})|^p \,\mathrm{d}x\right)^{1/p}, & 1 \le p < \infty, \\ \text{ess sup}_{\boldsymbol{x}\in\Omega} |f(\boldsymbol{x})|, & p = \infty \end{cases}$$

is finite. The space $C_c^\infty(\Omega)$, consisting of smooth functions which vanish outside a compact subset of Ω, is dense in $L^p(\Omega)$.

With a view to weak solutions, we also define the Sobolev spaces needed in this paper. Let $i \in \{1, 2, 3\}$ and note that if $f \in C^\infty(\Omega)$ and we denote $g_i = \partial_i f$, an integration by parts gives

$$\int_\Omega f(\boldsymbol{x})\partial_i\phi(\boldsymbol{x}) \,\mathrm{d}x = -\int_\Omega g_i(\boldsymbol{x})\phi(\boldsymbol{x}) \,\mathrm{d}x \quad \text{for all } \phi \in C_c^\infty(\Omega). \tag{1}$$

Now, whenever $f, g_i \in L^2(\Omega)$ satisfy (1), we say that g_i is the *weak ith partial derivative of* f and denote $\partial_i f = g_i$. This notation does not lead to confusion with classical partial derivatives since if f is continuously differentiable, the classical and weak partial derivatives coincide. The *Sobolev space* $H^1(\Omega)$ consists of those L^2-functions which possess L^2-integrable weak partial derivatives, and we endow $H^1(\Omega)$ with the norm

$$\|f\|_{H^1} \equiv (\|f\|_2 + \|\nabla f\|_2)^{1/2}.$$

The subspace H_0^1 is the closure of $C_c^\infty(\Omega)$ in the norm $\|\cdot\|_{H^1}$; equivalently, H_0^1 consists of those H^1-functions that vanish on the boundary $\partial\Omega$ in a trace sense.

Similarly, if $\boldsymbol{v} \in L^2(\Omega)^3$, we say that $\nabla\cdot\boldsymbol{v} = 0$ in the weak sense if $\int_\Omega \boldsymbol{v}(\boldsymbol{x})\cdot\nabla\phi(\boldsymbol{x}) \,\mathrm{d}x = 0$ for all $\phi \in C_c^\infty(\Omega)$. We define $L^2_\sigma(\Omega)$ as the closure of $C^\infty_{c,\sigma}(\Omega) \equiv \{\Phi \in C_c^\infty(\Omega)^3\colon \nabla\cdot\Phi = 0\}$ in $L^2(\Omega)^3$. Equivalently, $L^2_\sigma(\Omega) = \{\boldsymbol{v} \in L^2(\Omega)^3\colon \nabla\cdot\boldsymbol{v} = 0,\ \boldsymbol{v}\cdot\boldsymbol{n}|_{\partial\Omega} = 0\}$. We also denote $H^1_\sigma(\Omega) \equiv H^1(\Omega)^3 \cap L^2_\sigma(\Omega)$ and $H^1_{0,\sigma}(\Omega) \equiv H^1_0(\Omega) \cap L^2_\sigma(\Omega)$.

As we are studying evolutionary PDEs, we also need function spaces that depend on the time variable. When $f\colon \Omega \times [0, T) \to \mathbb{R}^d$ is a mapping, we denote by $f(t)$ the map $\boldsymbol{x} \mapsto f(\boldsymbol{x}, t)$. Given $p \in [1, \infty]$ and a separable Banach space X, the *Bochner space* $L^p(0, T; X)$ consists of those strongly measurable functions $f\colon (0, T) \to X$ for which

$$\|f\|_{L^p(0,T;X)} \equiv \begin{cases} \left(\int_0^T \|f(t)\|_X^p \,\mathrm{d}t\right)^{1/p}, & 1 \le p < \infty, \\ \text{ess sup}_{t\in(0,T)} \|f(t)\|_X, & p = \infty \end{cases}$$

is finite. In our applications, X is usually one of $L^2_\sigma(\Omega)$, $H^1(\Omega)^3$ and $H^1_0(\Omega)^3$. Note that if $\phi \in C_c^\infty(\Omega \times [0, T))$, it is allowed that $\phi(0) \not\equiv 0$; intervals $[0, \tau] \subset [0, T)$ are compact.

In the Bochner spaces of interest to us, bounded sequences do not, in general, have a convergence subsequence, but a weakly (weak-$*$) convergent subsequence can be found.

We illustrate this in the special case where X is a Hilbert space. We denote the inner product of X by $\langle \boldsymbol{v}, \boldsymbol{w}\rangle_X$. A sequence $(\boldsymbol{v}_j)_{j=1}^{\infty}$ converges weakly to $\boldsymbol{v}$ in $L^2(0,T;X)$ if $\int_0^T \langle \boldsymbol{v}_j, \psi\rangle_X \,\mathrm{d}t \to \int_0^T \langle \boldsymbol{v}, \psi\rangle_X \,\mathrm{d}t$ for every $\psi \in L^2(0,T;X)$.

2.2. Weak solutions and the ideal limit in MHD

We next give a definition of weak solutions of ideal MHD. It is, in fact, completely analogus to the notion of weak partial derivatives presented above. Suppose $(\boldsymbol{u}, \boldsymbol{B})$ form a smooth solution of the Cauchy problem for ideal MHD,

$$\partial_t \boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u} = -\nabla p + (\nabla\times\boldsymbol{B})\times\boldsymbol{B}, \qquad \nabla\cdot\boldsymbol{u} = 0, \tag{2}$$

$$\partial_t \boldsymbol{B} + \nabla\times(\boldsymbol{B}\times\boldsymbol{u}) = 0, \qquad \nabla\cdot\boldsymbol{B} = 0, \tag{3}$$

$$\boldsymbol{u}(\cdot,0) = \boldsymbol{u}_0, \qquad \boldsymbol{B}(\cdot,0) = \boldsymbol{B}_0 \tag{4}$$

in $\Omega\times[0,T)$, where $\Omega\subset\mathbb{R}^3$ is a magnetically closed domain Ω with an impenetrable boundary: $u_n = B_n = 0$ on S. Whenever $\phi\in C_c^\infty(\Omega\times[0,T))$ satisfies $\nabla\cdot\phi = 0$, an integration by parts and the formula $(\boldsymbol{u}\cdot\nabla)\boldsymbol{u} = \nabla\cdot(\boldsymbol{u}\otimes\boldsymbol{u})$ (valid since $\boldsymbol{u}$ is solenoidal) give

$$\int_0^T\int_\Omega [\boldsymbol{u}\cdot\partial_t\phi + (\boldsymbol{u}\otimes\boldsymbol{u} - \boldsymbol{B}\otimes\boldsymbol{B}) : \nabla\phi \,\mathrm{d}x\,\mathrm{d}t = -\int_\Omega \boldsymbol{u}_0\cdot\phi(\cdot,0)\,\mathrm{d}x \tag{5}$$

$$\int_0^T\int_\Omega [\boldsymbol{B}\cdot\partial_t\phi + (\boldsymbol{B}\times\boldsymbol{u})\cdot\nabla\times\phi]\,\mathrm{d}x\,\mathrm{d}t = -\int_\Omega \boldsymbol{u}_0\cdot\phi(\cdot,0)\,\mathrm{d}x. \tag{6}$$

Note that (3) and (4) do not involve derivatives of $\boldsymbol{u}$ and $\boldsymbol{B}$; in fact, they make sense even if $\boldsymbol{u}$ and $\boldsymbol{B}$ are merely square integrable. The notion of weak solution presented below is not the most general possible, but it is suitable for our purposes.

Definition 2.1: Suppose $\boldsymbol{u}_0, \boldsymbol{B}_0 \in L^2_\sigma(\Omega)$. A *weak solution* of ideal MHD is a couple $(\boldsymbol{u},\boldsymbol{B}) \in L^2(0,T;L^2_\sigma)$ that satisfies the weak formulation (3), (4).

Weak solutions of ideal MHD have turned out to be highly non-unique, and clearly numerous physically unrealistic solutions are admitted by the definition. One natural criterion for singling out physically relevant solutions is the ideal limit, which is defined below in analogy to the inviscid limit in hydrodynamics.

In viscous, resistive MHD, it is natural (and customary) to ask for more regularity than the mere square integrability of Definition 2.1. We use the non-slip, magnetic closedness and perfect conductivity boundary conditions, $\boldsymbol{u}|_{\partial\Omega} = \boldsymbol{0}$, $\boldsymbol{B}\cdot\boldsymbol{n}|_{\partial\Omega} = 0$ and $(\nabla\times\boldsymbol{B})\times\boldsymbol{n}|_{\partial\Omega} = \boldsymbol{0}$. By far the most commonly used notion of weak solution is that of *Leray–Hopf solutions*. They are couples $u\in L^\infty(0,T;L^2_\sigma)\cap L^2(0,T;H^1_{0,\sigma})$, $B\in L^\infty(0,T;L^2_\sigma)\cap L^2(0,T;H^1_\sigma)$ which satisfy a suitable analogue of (3), (4) (see Faraco and Lindberg 2020, p. 716) and the *energy inequality*

$$\tfrac{1}{2}(\|\boldsymbol{u}(t)\|_2^2 + \|\boldsymbol{B}(t)\|_2^2) + \int_0^t (\nu\|\nabla\boldsymbol{u}(\tau)\|_2^2 + \eta\|\nabla\boldsymbol{B}(\tau)\|_2^2)\,\mathrm{d}\tau \le \tfrac{1}{2}(\|\boldsymbol{u}(0)\|_2^2 + \|\boldsymbol{B}(0)\|_2^2) \tag{7}$$

for all $t \in [0, T)$. (Often, a so-called strong energy inequality is assumed instead of (5), but this distinction is irrelevant for our purposes.)

In the case of the Navier–Stokes equations, the existence of Leray–Hopf solutions was proved by Leray in 1934 on the whole space $\mathbb{R}^3$ and by Hopf in 1951 on all domains of $\mathbb{R}^3$. For the existence of Leray–Hopf solutions of viscous, resistive MHD see Sermange and Temam (1983) in simply connected domains and the torus $\mathbb{T}^3$ as well as Faraco and Lindberg (2020) in multiply connected domains. We are now in a position to define the ideal limit in direct analogy to the inviscid limit in hydrodynamics.

Definition 2.2: Suppose that viscosities $\nu_j > 0$ and resistivities $\eta_j > 0$ satisfy $\nu_j, \eta_j \to 0$ and that $\boldsymbol{u}_{0,j} \to \boldsymbol{u}_0$ and $\boldsymbol{B}_{0,j} \to \boldsymbol{B}_0$ in $L^2_\sigma(\Omega)$. Assume that for every $j \in \mathbb{N}$, $(\boldsymbol{u}_j, \boldsymbol{B}_j)$ is a Leray–Hopf solution of viscous, resistive MHD with initial datum $(\boldsymbol{u}_{0,j}, \boldsymbol{B}_{0,j})$ and that $\boldsymbol{u}, \boldsymbol{B} \in L^\infty(0, T; L^2_\sigma)$. We say that $(\boldsymbol{u}, \boldsymbol{B})$ is the *ideal limit* or inviscid, non-resistive limit of $(\boldsymbol{u}_j, \boldsymbol{B}_j)$ (in the energy norm) if $\|\boldsymbol{u}_j - \boldsymbol{u}\|_{L^\infty(0,T;L^2)} \to 0$ and $\|\boldsymbol{B}_j - \boldsymbol{B}\|_{L^\infty(0,T;L^2)} \to 0$. We then also say that the ideal limit holds for $(\boldsymbol{u}_j, \boldsymbol{B}_j)$ and $(\boldsymbol{u}, \boldsymbol{B})$.

It is immediate from (3), (4) that ideal limits $(\boldsymbol{u}, \boldsymbol{B})$ satisfy the ideal MHD equations. It is a major open problem under which conditions the ideal limit holds (see, e.g. Wu 1997, Xiao *et al.* 2009). In hydrodynamics, a thorough recent review on the inviscid limit is given by Maekawa and Mazzucato (2018).

3. Conserved and dissipated quantities at the ideal limit

We are now in a position to formulate the main results mentioned in the introduction. We also recall other related mathematical results on classical conserved quantities of ideal MHD. The quantities covered include total energy, cross helicity, magnetic helicity (in 3D) and mean-square magnetic potential (in 2D). In fact, every regular Casimir in 3D ideal MHD is a function of magnetic helicity and cross helicity (Khesin *et al.* 2021). In hydrodynamics, a similar result holds for helicity (Enciso *et al.* 2016).

3.1. Magnetic helicity

In this section, we review rigorous results on the conservation of magnetic helicity by weak solutions of ideal MHD. First, however, we discuss the definition of magnetic helicity. If Ω is either a bounded, simply connected and magnetically closed domain or the torus $\mathbb{T}^3$, the definition $\mathcal{H}(B)(t) \equiv \int_\Omega \boldsymbol{A}(\boldsymbol{x}, t) \cdot \boldsymbol{B}(\boldsymbol{x}, t)\, \mathrm{d}x$ for $\boldsymbol{B} \in L^2(0, T; L^2_\sigma)$ and $\boldsymbol{A} \in L^2(0, T; W^{1,2})$ with $\nabla \times \boldsymbol{A} = \boldsymbol{B}$ is gauge-invariant. In the more physically relevant case of multiply connected domains, however, the orthogonal complement of L^2_σ strictly contains the kernel of curl, which leads to the gauge dependence of $\int_\Omega \boldsymbol{A}(\boldsymbol{x}, t) \cdot \boldsymbol{B}(\boldsymbol{x}, t)\, \mathrm{d}x$. In a toroidal domain, the classical Bevir–Gray formula (see MacTaggart and Valli 2019, p. 2) gives a gauge-invariant modification of classical magnetic helicity.

In multiply connected bounded domains with a Lipschitz-continuous boundary, a gauge-invariant definition of magnetic helicity was recently given by MacTaggart and Valli (2019). If the first Betti number of Ω is $g > 0$, MacTaggart and Valli consider $2g$ non-bounding cycles on $\partial\Omega$, $\{\Gamma_j\}_{j=1}^g \cup \{\Gamma_j'\}_{j=1}^g$, that represent the generators of the first homology group of $\partial\Omega$. In Ω there are g cutting surfaces Σ_j which satisfy $\Sigma_j \subset \Omega$ and

$\partial \Sigma_j = \Gamma_j'$, and in addition, each Σ_j cuts the cycle Γ_j. The gauge-invariant definition of magnetic helicity is given by $\Upsilon(\boldsymbol{B}) \equiv \int_\Omega \boldsymbol{A} \cdot \boldsymbol{B} - \sum_{j=1}^{g} (\oint_{\Gamma_j} \boldsymbol{A} \cdot \boldsymbol{t}_j)(\int_{\Sigma_j} \boldsymbol{B} \cdot \boldsymbol{n}_j)$. The MacTaggart–Valli formula reduces to the Bevir–Gray formula on the torus and to classical magnetic helicity in the simply connected case. Magnetic helicity is conserved by smooth solutions of ideal MHD in multiply connected domains, which can be seen by following the calculations in Faraco *et al.* (2022).

The first result on magnetic helicity conservation for weak solutions was given by Caflisch *et al.* (1997): conservation holds in certain Besov spaces. Conservation was shown for solutions L^3-integrable in space and time, independently by Eyink and Aluie (see Aluie 2009) and by Kang and Lee (2007). Nevertheless, a rigorous proof of (the mathematical version of) Taylor's conjecture requires conservation for solutions that arise at the ideal limit, and for them only square integrability in space is available. The proofs of the L^3 result break down, but one may instead exploit the fact that Leray–Hopf solutions satisfy the energy inequality. The mathematical version of Taylor's conjecture was proved by Faraco and Lindberg (2020) in simply connected domains and by Faraco *et al.* (2022) in multiply connected domains:

Theorem 3.1: *Magnetic helicity is conserved at the ideal limit. More generally, all weak L^2-limits of Leray–Hopf solutions, where $\nu, \eta \to 0$, conserve magnetic helicity.*

Three variants of Theorem 3.1 have appeared in fluid dynamics: in 2D Euler, energy is conserved at the inviscid limit (Cheskidov *et al.* 2016), in SQG, the Hamiltonian is conserved at the inviscid limit (Constantin *et al.* 2018), and in 2D MHD, the mean-square magnetic potential is conserved at the ideal limit (Faraco and Lindberg 2020). All four phenomena seem to fall under the scope of the celebrated *self-organisation conjecture* described below.

Mathematically, self-organisation concerns a dissipative nonlinear PDE and involves two quadratic or higher-order integral quantities $A(t)$ and $B(t)$ with the following properties (Hasegawa 1985, p. 2): in the inviscid case, $A(t)$ and $B(t)$ are conserved by smooth solutions; in the presence of slight viscocity, $A(t)$ is dissipated much faster than $B(t)$; and $A(t)$ undergoes a forward cascade whereas $B(t)$ undergoes an inverse cascade. The self-organisation conjecture then predicts that, under slight dissipation, the system evolves towards a state where $A(t)$ is minimised under the constraint that $B(t) \approx B(0)$ is given. In Woltjer–Taylor relaxation theory, $A(t)$ is the total energy and $B(t)$ is the magnetic helicity. In 2D MHD, $B(t)$ is the mean-square magnetic potential instead.

It is, however, acknowledged that the predictions of Woltjer–Taylor relaxation theory and the self-organisation conjecture need to be modified if there are more topological invariants that survive the relaxation (see, e.g. Bhattacharjee and Dewar 1982, Moffatt 1985, Arnold and Khesin 1998, Yeates *et al.* 2010, Komendarczyk 2022). The mathematical analysis of many of these invariants in the context of weak solutions is complicated by the issue of spontaneous stochasticity; not only the Lagrangian particle trajectories but also the magnetic field lines cease to be well-defined in the classical sense, even in the Hölder continuous regime (see, e.g. Lazarian *et al.* 2020, pp. 14–15) and the references contained therein.

For general weak solutions of ideal MHD (which need not arise at the ideal limit), Beekie *et al.* (2020) showed that magnetic helicity conservation fails already at the regularity level

$C([0,T);H^{\beta}) \subset C([0,T);L^{2+\epsilon})$, where $0 < \beta, \epsilon \ll 1$. Buckmaster and Vicol (2021) presented the natural conjecture that whenever $2 \leq p < 3$, there exist weak solutions $\boldsymbol{u}, \boldsymbol{B} \in L^3(0,T;L^p(\mathbb{T}^3)^3)$ of ideal MHD that do not conserve magnetic helicity. A stronger version of the conjecture was recently proved by Faraco *et al.* (2021b):

Theorem 3.2: *There exist weak solutions $\boldsymbol{u}, \boldsymbol{B} \in L^{\infty}(0,T;L^{3,\infty}(\mathbb{T}^3)^3)$ of ideal MHD that do not conserve magnetic helicity in time.*

All the theorems mentioned above concern either periodic solutions of MHD or magnetically closed domains. In astrophysical applications, however, typically the magnetic field threads the boundaries of interest. In magnetically open (simply connected) domains, one uses the *relative magnetic helicity* of Berger and Field (1984), Jensen and Chu (1984) and Finn and Antonsen (1985) instead. By solving the Neumann problem $-\Delta g = 0$, $(\partial g/\partial \boldsymbol{n} - \boldsymbol{B}\cdot\boldsymbol{n})|_{\partial\Omega} = 0$ one obtains a potential field $\boldsymbol{B}_P \equiv \nabla g$ with the properties $\nabla\cdot\boldsymbol{B}_P = 0$, $\nabla\times\boldsymbol{B}_P = 0$ and $(\boldsymbol{B}_P - \boldsymbol{B})\cdot\boldsymbol{n}|_{\partial\Omega} = 0$. The relative magnetic helicity of $\boldsymbol{B}$ at time t is given by the *Finn–Antonsen formula* $\mathcal{H}_R(\boldsymbol{B}) \equiv \int_{\Omega}(\boldsymbol{B}-\boldsymbol{B}_P)\cdot(\boldsymbol{A}+\boldsymbol{A}_P)\,\mathrm{d}x$, where $\nabla\times\boldsymbol{A}_P = \boldsymbol{B}_P$. It is manifestly gauge invariant in both $\boldsymbol{A}$ and $\boldsymbol{A}_P$. It is an intriguing problem to decide whether, in the ideal limit, $\mathcal{H}_R(\boldsymbol{B})$ is a conserved quantity (in the sense that its time evolution only depends on boundary fluxes, in some natural collection of gauges).

3.2. Total energy and cross helicity

We move on to mathematical results on energy decay in ideal MHD. Taken literally, the possibility of energy decay already follows from work on Euler equations since ideal MHD reduces to Euler when $\boldsymbol{B} = 0$. For non-trivial magnetic fields, the first examples were constructed by Bronzi *et al.* (2015) using the machinery of De Lellis and Székelyhidi (2009). They constructed bounded solutions of the "$2\frac{1}{2}$-dimensional" form $\boldsymbol{u}(\boldsymbol{x},t) = (u_1(x_2,x_2,t), u_2(x_1,x_2,t), 0)$ and $\boldsymbol{B}(\boldsymbol{x},t) = (0,0,b_3(x_1,x_2,t))$ which violate energy conservation. The magnetic helicity and cross helicity of the solutions trivially vanish.

The first truly three-dimensional energy dissipative solutions were constructed independently by Beekie *et al.* (2020) and Faraco *et al.* (2021a). In the former result, magnetic helicity, total energy and cross helicity all vary in time, whereas the latter is in the integrability regime L^{∞} where magnetic helicity is automatically conserved. We present a refinement of the main result of Faraco *et al.* (2021a) from Faraco *et al.* (2021b) which allows us to prescribe the values of total energy and cross helicity. In the statement, we denote total energy by $\mathcal{E}(t)$ and cross helicity by $\mathcal{W}(t)$.

Theorem 3.3: *There exists a constant $M > 0$ with the following property. Let $h \in \mathbb{R}$ and suppose $e, w \in C([0,T])$ with $e(t) - |w(t)| > M|h|$ for all t. Then there exists a weak solution $(\boldsymbol{u},\boldsymbol{B}) \in L^{\infty}(0,T;L^{\infty}(\mathbb{T}^3)^3)$ of (2)–(4) such that*

$$\mathcal{E}(\boldsymbol{u},\boldsymbol{B})(t) = e(t), \quad \mathcal{W}(\boldsymbol{u},\boldsymbol{B})(t) = w(t), \quad \mathcal{H}(\boldsymbol{B})(t) = h \quad \text{for a.e. } t.$$

In Theorem 3.3, selective decay of energy is observed on weak solutions of ideal MHD, as is seen by choosing the prescribed energy $e(t)$ to be a decreasing function of time. The

theorem gives some rigorous support for the contention that there is nothing inherent in the ideal MHD system prohibiting magnetic reconnection. As expectable, the theory of weak solutions to MHD faces similar obstructions and desires as Euler or Navier–Stokes. An important aspect would be to obtain the results as ideal limits. In hydrodynamic turbulence, the use of the inviscid limit as a selection criterion has been advocated, e.g. by Eyink (2018). We also mention that Bardos *et al.* (2012) have analysed the inviscid limit as a selection criterion when the initial datum is a shear flow.

An aspect that seems more tractable in the near future is adapting Onsager's conjecture to MHD turbulence and obtaining Hölder continuous solutions. The naturality of the class $C^{1/3}$, from a physical point of view, is discussed in the references Aluie (2017), Dubrulle (2019) and Eyink (2018) among many others. Relatedly, from a mathematical viewpoint, $C^{1/3}$ is the conjectured regularity threshold for the conservation of total energy and cross helicity (see Buckmaster and Vicol 2021).

In any case, it seems to us that lack of uniqueness of irregular weak solutions is in agreement with the natural stochasticity observed in turbulent regimes. There is no hope to restore uniqueness at such fine scales, but there should at least be a selection criterion for picking a unique subsolution (coarse-grained solution); see Otto (1999) and Székelyhidi (2012) for such a selection criterion in the context of incompressible porous media equations.

3.3. Mean-square magnetic potential (2D MHD)

In 2D ideal MHD, magnetic helicity is replaced by a scalar quantity, the *mean-square magnetic potential* $\int_\Omega |\psi|^2 \,\mathrm{d}x$, where $\nabla^\perp \psi = \boldsymbol{B}$. A canonical choice of ψ depends on the geometry of Ω; on $\mathbb{T}^2$, the mean $\langle \psi \rangle$ is chosen to vanish at all times. For the conservation of mean-square magnetic potential in suitable Besov spaces (see Caflisch *et al.* 1997). Conservation in the energy space $L^\infty(0, T; L^2(\mathbb{T}^2)^2)$ was shown by Faraco and Lindberg (2018). In fact, Faraco and Lindberg (2020) presented a refined result where we suppose $\Omega \subset \mathbb{R}^2$ is a bounded domain whose boundary is Lipschitz-continuous and has finitely many components.

Theorem 3.4: *The mean-square magnetic potential is conserved in time by solutions* $\boldsymbol{u}, \boldsymbol{B} \in L^\infty(0, T; L^2_\sigma)$ *of ideal MHD as well as weak limits of Leray–Hopf solutions when* $\eta, \nu \to 0$.

We present the main ideas of the proof, working on $\mathbb{T}^2$ for simplicity. We introduce a zero-mean stream function: $\nabla^\perp \phi = \boldsymbol{u}$, so that $\partial_t \psi + J(\phi, \psi) = 0$, where $J(\phi, \psi) = \partial_1 \phi \partial_2 \psi - \partial_2 \phi \partial_1 \psi$ is the Jacobian determinant. If $\boldsymbol{u}$ and $\boldsymbol{B}$ are smooth, the conservation of the mean-square magnetic potential is obtained by an integration by parts:

$$\partial_t \int_{\mathbb{T}^2} |\psi|^2 \,\mathrm{d}x = -2 \int_{\mathbb{T}^2} \psi J(\phi, \psi) \,\mathrm{d}x = 2 \int_{\mathbb{T}^2} \phi J(\psi, \psi) \,\mathrm{d}x = 0. \tag{8}$$

The underlying reason for the fact that conservation extends to weak solutions $\boldsymbol{u}, \boldsymbol{B} \in L^\infty(0, T; L^2(\mathbb{T}^2)^2)$ is the *higher integrability* of the Jacobian determinant, as explained below. The proof is representative of arguments concerning conservation laws for weak solutions; conservation of magnetic helicity for L^3-solutions of ideal 3D MHD is obtained in a very similar manner.

Proof: The time derivative of $\int_{\mathbb{T}^2} |\psi|^2 \, \mathrm{d}x$ is difficult to handle directly, and so we use its weak definition and attempt to show that $\int_0^T \eta'(t) \int_{\mathbb{T}^2} |\psi(x,t)|^2 \, \mathrm{d}x \, \mathrm{d}t = 0$ for all $\eta \in C_c^\infty(0,T)$. By basic properties of space-time mollifiers $\psi_\delta \equiv \psi * G_\delta$, where G is a standard mollification kernel, and an integration by parts we get

$$\begin{aligned}\int_0^T \eta'(t) \int_{\mathbb{T}^2} |\psi(x,t)|^2 \, \mathrm{d}x \, \mathrm{d}t &= \lim_{\delta \to 0} \int_0^T \eta'(t) \int_{\mathbb{T}^2} |\psi_\delta(x,t)|^2 \, \mathrm{d}x \, \mathrm{d}t \\ &= \lim_{\delta \to 0} 2 \int_0^T \eta(t) \int_{\mathbb{T}^2} \psi_\delta(x,t) [J(\phi,\psi)]_\delta \, \mathrm{d}x \, \mathrm{d}t.\end{aligned}$$

By decomposing $[J(\phi,\psi)]_\delta = [J(\phi - \phi_\delta, \psi)]_\delta + ([J(\phi_\delta,\psi)]_\delta - J(\phi_\delta,\psi)) + J(\phi_\delta,\psi)$ we end up with three double integrals, and we would like each of them to vanish at the limit $\delta \to 0$. Here, the higher integrability of J comes to the rescue. Below, we focus on $\int_{\mathbb{T}^2} \phi_\delta(x,t)[J(\phi - \psi_\delta, \psi)]_\delta \, \mathrm{d}x$ and the other two spatial integrals at each time slice.

Whenever $f, g, h \in W^{1,2}(\mathbb{T}^2)$, the Cauchy–Schwarz inequality readily yields $J(g,h) \in L^1(\mathbb{T}^2)$, but in fact, $J(g,h)$ is locally $L \log L$-integrable in subdomains where it is non-negative (Müller 1989). More precisely, $J(g,h)$ belongs to the *real-variable Hardy space* $\mathcal{H}^1(\mathbb{T}^2)$ (Coifman *et al.* 1993) whose dual is $\mathrm{BMO}(\mathbb{T}^2)$ (Fefferman and Stein 1972). The integral $\int_{\mathbb{T}^2} f J(g,h) \, \mathrm{d}x$ can then be understood rigorously in terms of so-called $\mathcal{H}^1 - \mathrm{BMO}$ duality. One estimates

$$\left| \int_{\mathbb{T}^2} f J(g,h) \, \mathrm{d}x \right| \lesssim \|f\|_{\mathrm{BMO}} \|J(g,h)\|_{\mathcal{H}^1} \lesssim \|\nabla f\|_2 \|\nabla g\|_2 \|\nabla h\|_2,$$

recovering the beautiful *Wente inequality*. This allows one to show that $\lim_{\delta \to 0} \int_0^T \eta(t) \int_{\mathbb{T}^2} \psi_\delta(x,t)[J(\phi - \phi_\delta, \psi)]_\delta \, \mathrm{d}x \, \mathrm{d}t \to 0$, and a similar argument applies to the other two integrals. We have thereby turned the formal computation (6) into a rigorous argument at the level of weak solutions $\boldsymbol{u}, \boldsymbol{B} \in L^\infty(0,T; L^2_\sigma(\mathbb{T}^2)^2)$. ■

The higher integrability of J is sometimes called *compensated integrability*. A fact that has intrigued mathematicians for the last 30 years (see Coifman *et al.* 1993, p. 267) is that there seems to be a tight connection between compensated integrability and compensated compactness (weak continuity), the key concept in the forthcoming section 4.1. The weak continuity of Jacobians is a classical result that goes back at least to Morrey (1966) and Reshetnyak (1968). In fact, compensated integrability and compensated compactness are equivalent for a large class of nonlinearities (see Guerra and Raita 2019, Guerra *et al.* 2020).

4. The convex integration method

Convex integration, a terminology borrowed from differential geometry, is a programme to obtain unexpected solutions in a variety of problems, calculus of variations or elliptic partial differential equations for example. De Lellis and Székelyhidi discovered that in combination with Tartar compensated compactness, the method was amenable to treat the Euler equations, and indeed many other models of hydrodynamics. As it happens, the method is based on wrapping the actual PDE of interest within a (more general) macroscopic version which is of its own interest. Then from the macroscopic solutions an actual

weak solution is built. In the first subsection, we describe such macroscopic solutions, the relaxation, and in the second the convex integration method and its consequences for magnetohydrodynamics.

4.1. Relaxation and Tartar framework

In the context of nonlinear PDEs, mathematical relaxation appears naturally through the concept of weak limits. In the 1970–80s L. Tartar developed a far-reaching framework to treat various nonlinear partial differential equations arising in continuum physics (Tartar 1979, 1983). The key feature of Tartar's approach is to treat separately constitutive relations, expressed as pointwise constraints, and the linear differential equations arising from conservation laws. In this context, weak limits can be interpreted as deterministic analogues of various averaging processes (similar to coarse-graining or low-pass filtering) without specifying a particular length scale (see also DiPerna 1985, Lax 1991 and Bardos *et al.* 2000 for related ideas). Notice that weak limits consider functions as distributions and hence are particularly suitable to deal with weak (distributional) solutions.

Next, we will explain the basic idea, and illustrate it with the Magnetic part of MHD for simplicity.

In short, within the Tartar framework one decouples a nonlinear partial differential equation into a conservation law and a nonlinear constraint:

$$\mathcal{L}(U) = 0, \qquad U(\boldsymbol{x}, t) \in \mathcal{K}. \tag{9}$$

Here $U = (u^1, u^2, \ldots, u^m) : \Omega \times [0, T] \to \mathbb{R}^m$ represents augmented states, $\mathcal{K} \subset \mathbb{R}^m$ is a set, which we keep independent of $(\boldsymbol{x}, t)$ for simplicity, and $\mathcal{L}$ is a linear differential operator $\mathcal{L}(U) = \sum_{i,j} A^i_{j,k} \partial_i u_j$. In the constraint set $\mathcal{K}$ one can encode further information such as prescribing energy densities.

Neglecting the conservation of momentum in MHD, for the time being, we define $U = (\boldsymbol{E}, \boldsymbol{B}, \boldsymbol{u})$, i.e. we have restored the electric field. In this context, by writing $\mathcal{L}(U) = (\partial_t \boldsymbol{B} + \nabla \times \boldsymbol{E}, \operatorname{div}(\boldsymbol{u}), \operatorname{div}(\boldsymbol{B}))$ and $\mathcal{K} = \{(\boldsymbol{E}, \boldsymbol{B}, \boldsymbol{u}) \in (\mathbb{R}^3)^3 : \boldsymbol{E} = \boldsymbol{B} \times \boldsymbol{u}\}$ one recovers the *non-resistive kinematic dynamo equations* (Arnold and Khesin 1998, p. 259).

In this formulation, for sequences U_j weakly converging to U and satisfiying $\mathcal{L}(U_j) = 0$ and $U_j(x, t) \in \mathcal{K}$ a.e. (x, t), the weak limits still satisfy that $\mathcal{L}(U) = 0$ but there exists a larger set $\operatorname{Relax}(\mathcal{K})$ such that $U(x, t) \in \operatorname{Relax}(\mathcal{K})$ a.e. (x, t). (We choose $\operatorname{Relax}(\mathcal{K})$ to be minimal with respect to this property.)

From this perspective, coarse-grained solutions, mean-field equations or other ways to describe macroscopic variables are replaced by the understanding of solutions (with the correct integrability assumption) $\bar{U}$ of

$$\mathcal{L}(\bar{U}) = 0, \qquad \bar{U} \in \operatorname{Relax}(\mathcal{K}). \tag{10}$$

In the nomenclature of convex integration theory for hydrodynamics, such solutions are called *subsolutions*.

The advantage of Tartar relaxation with respect to other averaging processes is that on the one hand it deals at once with all length scales and on the other hand it provides geometric insights which remain hidden in other approaches.

For example, by the classical lower semicontinuity of functionals with convex densities, we have $\mathrm{Relax}(\mathcal{K}) \subset \mathrm{conv}(\mathcal{K})$, where $\mathrm{conv}(\mathcal{K})$ is the classical convex hull of $\mathcal{K}$. Tartar Compensated Compactness theory provides us with a wealth of lower and weakly semicontinuous functionals in addition to the convex ones, when acting on solutions to (7), and gives a geometric approach to them. Let us define the *wave cone* Λ as the set of directions in which a plane wave is a solution to (7):

$$\begin{aligned}\Lambda = \{U \in \mathbb{R}^m :\ &\text{there exists } (\xi, \xi_t) \in \mathbb{R}^d \times \mathbb{R} \text{ such that for every } h : \mathbb{R} \to \mathbb{R} \\ &\mathcal{L}(h(\langle (x,t), (\xi, \xi_t)\rangle)) = 0\}\end{aligned} \tag{11}$$

A function $f : \mathbb{R}^m \to \mathbb{R}$ is said to be Λ-*convex* if $f(U + tV)$ is a convex function of t whenever $V \in \Lambda$. Now, Λ-convexity is a necessary condition for the lower semicontinuity for densities f of integral functionals $\int_{\Omega\times[0,T]} f(U(\boldsymbol{x}, t))\,\mathrm{d}x\,\mathrm{d}t$ acting on the kernel of $\mathcal{L}$. Tartar (1979) proved that this condition is also sufficient for quadratic functionals. A very famous problem in the area asks in which contexts this condition is also sufficient; Morrey's problem is the crucial special case of rank-one convexity. In any case, when such nonlinearities f are weakly continuous we call them *compensated compactness quantities*.

Therefore, if $\mathcal{K}^\Lambda$ stands for the Λ-convex hull (the set of points that cannot be separated from $\mathcal{K}$ via Λ-convex functions) and, loosely speaking, $\mathcal{K}^{Cc}$ takes into account compensated compactness quantities, one obtains the following geometric insight on the relaxation:

$$\mathcal{K}^\Lambda \subset \mathrm{Relax}(\mathcal{K}) \subset \mathcal{K}^{Cc}.$$

As we will see in the next section, it follows from De Lellis–Székelyhidi theory that the relaxation of the Euler equation can be understood through standard convexity. However, MHD hides an appealing surprise connected to the conservation of magnetic helicity.

Indeed, Tartar himself discovered that by treating the Faraday–Maxwell system separately (Maxwell equations in vacuum), $\boldsymbol{B} \cdot \boldsymbol{E}$ is a weakly continuous quantity when $\boldsymbol{B}$ and $\boldsymbol{E}$ have the natural integrability conditions. A conceptually pleasing way to see this is to recast such equations using the *Faraday 2-form* $\omega \in \Lambda(\mathbb{R}^4)$,

$$\omega = \epsilon_{ijk} B_i\,\mathrm{d}x_j \wedge \mathrm{d}x_k + E_i\,\mathrm{d}x_i \wedge \mathrm{d}t,$$

where ϵ_{ijk} are permutation symbols. Thus the homogeneous Maxwell equations read simply as $\mathrm{d}\omega = 0$, and the Poincaré lemma furnishes a 1-form α (the electromagnetic potential) such that $\mathrm{d}\alpha = \omega$. It is easy to see that $\omega \wedge \omega = \mathrm{d}(\alpha \wedge \omega)$ is weakly continuous, and since

$$\omega \wedge \omega = \boldsymbol{B} \cdot \boldsymbol{E}\,\mathrm{d}x_1 \wedge \mathrm{d}x_2 \wedge \mathrm{d}x_3 \wedge \mathrm{d}t,$$

we have proved the weak continuity of $\boldsymbol{B} \cdot \boldsymbol{E}$. All this analysis is related to the linear operator $\mathcal{L}$, but we observe that if $U(x,t) \in \mathcal{K}$ a.e., then $\boldsymbol{E} = \boldsymbol{B} \times \boldsymbol{u}$. Therefore $\boldsymbol{B} \cdot \boldsymbol{E} = 0$, and by weak continuity, the same is true in the relaxation.

The relaxation of the ideal MHD system must satisfy the additional pointwise constraint

$$\boldsymbol{B} \cdot \boldsymbol{E} = 0, \tag{12}$$

up to a set of zero measure.

We next discuss the crucial role that the compensated compactness quantity $\boldsymbol{B} \cdot \boldsymbol{E}$ plays in the conservation of magnetic helicity. Working again on the torus for simplicity, and not invoking the ideal Ohm's law $\boldsymbol{E} = \boldsymbol{B} \times \boldsymbol{u}$ just yet, a classical computation gives

$$\partial_t \int_{\mathbb{T}^3} \boldsymbol{A}(x,t) \cdot \boldsymbol{B}(\boldsymbol{x},t)\,\mathrm{d}x = -2 \int_{\mathbb{T}^3} \boldsymbol{B}(x,t) \cdot \boldsymbol{E}(\boldsymbol{x},t)\,\mathrm{d}x \tag{13}$$

for all smooth solutions $(\boldsymbol{u}, \boldsymbol{B})$. Just like we turned the formal computation (6) into a rigorous argument in the proof of Theorem 3.4, (11) extends to weak solutions as long as, e.g. $\boldsymbol{B} \in L^3(\mathbb{T}^3 \times (0,T))$ and $\boldsymbol{E} \in L^{3/2}(\mathbb{T}^3 \times (0,T))$. This is just enough integrability to make $\boldsymbol{B} \cdot \boldsymbol{E}$ integrable via Hölder's inequality: $\int_0^T \int_{\mathbb{T}^3} |\boldsymbol{B} \cdot \boldsymbol{E}|\,\mathrm{d}x\,\mathrm{d}t \leq \|\boldsymbol{B}\|_{L^3} \|\boldsymbol{E}\|_{L^{3/2}}$. This is the basic reason behind the L^3 result of Aluie and Eyink and Kang and Lee; note that the L^3-integrability of $\boldsymbol{u}$ and $\boldsymbol{B}$ yields $\boldsymbol{B} \times \boldsymbol{u} \in L^{3/2}(\mathbb{T}^3 \times (0,T))$. In view of (10)–(11), magnetic helicity conservation also holds at the level of the relaxed equation under proper integrability assumptions. This is yet another manifestation of the robustness of magnetic helicity conservation.

4.2. Convex integration

Mathematical relaxation has appeared in various guises since the Tartar framework, in the theory of homogenisation (Milton 2002) or in variational models in physics (Ball and James 1987). In the mid-90s, Müller and Sverák (1996) (see also Müller and Sverák 2003) realised that in the context of gradient fields it could be combined with the theory of Nash (1954) and Kuiper (1955), largely developed by Gromov (1986), to provide unexpected solutions to a number of problems. See Kirchheim *et al.* (2003) for a nice descripition of the method.

De Lellis and Székelyhidi (2009) made a revolution in the theory of weak solutions in hydrodynamics combining convex integration with the Tartar framework explained in the previous chapter to provide unexpected solutions to the Euler equations. De Lellis and Székelyhidi not only recast former results of Scheffer (1993) and Shnirelman (1997) concerning lack of uniqueness or dissipation of energy in a much more robust way but they opened a Pandora's box of wild solutions to a variety of problems in hydrodynamics.

The research in the area has been extremely intense since the first paper and by now there is a solid theory (see, e.g. the reviews De Lellis and Székelyhidi 2012, 2017, 2022). We refrain from detailing all the achievements of the method concerning the Euler equation but we mention the solution of Onsager's conjecture (Isett 2018, Buckmaster *et al.* 2019) culminating the ideas by many mathematicians, e.g. De Lellis and Székelyhidi (2009, 2013, 2014), Buckmaster (2015), Daneri and Székelyhidi (2017) or non-uniqueness of weak solutions to Navier–Stokes (Buckmaster and Vicol 2019a); on the latter also see, e.g. Buckmaster *et al.* (2018), Cheskidov and Luo (2020, 2021) and Buckmaster and Vicol (2021).

Since we want to discuss convex integration in the context of relaxation, let us remark that the following terminology from the theory of isometric immersions is often used in this context.

H-Principle: If there exists a subsolution, then there are infinitely many solutions to the original problem which behave macroscopically as the subsolution, i.e. sharing initial data.

Therefore, even though it is likely that in turbulent regimes there is some inherent stochasticity for the true solutions, leading to the lack of uniqueness for deterministic solutions, the main macroscopic features can be read from the subsolution.

In any case, the method of De Lellis and Székelyhidi has shown to be flexible enough to deal with numerous other equations in hydrodynamics (see, e.g.Córdoba *et al.* 2011, Shvydkoy 2011, Székelyhidi 2012, Chiodaroli *et al.* 2015, Modena and Székelyhidi 2018, Buckmaster and Vicol 2019b, Mengual 2020). Particulary, the previously unaccessible realm of modelling instabilities in fluid dynamics has been shown amenable to be treated by the method.

An achievement of the method has been providing solutions dealing with Saffman–Taylor, Rayleigh–Taylor and Kelvin–Helmholtz instabilities (see Székelyhidi 2012, Förster and Székelyhidi 2018, Castro *et al.* 2019, 2021a, 2021b, Arnaiz *et al.* 2021, Gebhard and Kolumbán 2021, 2022, Gebhard *et al.* 2021, Mengual and Székelyhidi 2022). It is remarkable that the h-principle philosophy explained above is particularly pleasant in these models and the dissipation of energy and size and shapes of various mixing zones can be read from the subsolutions.

In short, convex integration reverses the process of the relaxation. The relaxation of the Euler equation can be read as admitting a positive definite Reynolds stress $\mathbf{R}$. Thus in the relaxation, the momentum equation takes the form

$$\partial_t \bar{\boldsymbol{u}} + \operatorname{div}(\bar{\boldsymbol{u}} \otimes \bar{\boldsymbol{u}}) + \nabla \bar{p} = -\operatorname{div}(\bar{\mathbf{R}}).$$

Such a relaxed solution $\bar{U} = (\bar{\boldsymbol{u}}, \bar{p}, \bar{\mathbf{R}})$ was obtained as a complicated weak limit, or a combination of coarse-grained solutions, perhaps taking into account multiple scales. All manifestations of convex integration rely on an iterative process consisting in adding localised oscillating approximate solutions tuned to a particular length scale (i.e. suitable coarse-grained solutions) to yield back a weak solution. That is, from a subsolution $\bar{U}$ one generates a solution U,

$$U = \bar{U} + \sum_q U_q,$$

where U_q is an oscillating solution of the conservation law at a certain scale indexed by q (and, in the more advanced recent papers, also concentrated).

The iterative process is carefully crafted so the iterations never leave the set $\mathrm{Relax}(\mathcal{K})$ but converge to the set $\mathcal{K}$ in the limit. In the Euler case $\mathcal{K} = \{(\boldsymbol{u}, p, \mathbf{R}) : \mathbf{R} = 0\}$ and $\mathrm{Relax}(\mathcal{K}) = \{(\boldsymbol{u}, p, \mathbf{R}) : \mathbf{R} \geq 0\}$

Indeed, the very idea appeared already in the original paper of Nash (1954) on constructing C^1 isometric embeddings. Tartar plane waves furnished existence of solutions, and in order to achieve more regular solutions, selecting or inventing the approximate solutions and gluing them properly to optimise the length scales and consequently, the regularity of the final solution has become really a work of art, related for example with the well-known intermittency phenomenon in turbulence.

In any case, in order to get started with the iterative scheme, it is much easier if the set $\mathrm{Relax}(\mathcal{K})$ has a non-empty interior, e.g. in the Euler situation, this reduces to the Reynolds stress being positive definite and additional considerations. This openness condition allows to produce small errors at step q which could be taken care of in the step $q+1$. The original iterations U_q were based on Tartar plane waves, and subsequently, new master

functions such as classical Beltrami flows, or ad hoc discovered Mikado flows (Daneri and Székelyhidi 2017) were added to obtain more regular solutions.

Let us warn the reader that the enormous success of the method of convex integration has its limitations. A frontal one is that there exist "rigid" equations for which the relaxation (in the proper function spaces) coincides exactly with the original equation. This is in stark contrast to "flexible" models, such as the Euler equations, where Relax($\mathcal{K}$) is large (with a non-empty interior). MHD is structurally different from both rigid and flexible equations, as we discuss below.

Recall from section 4.1 that $\boldsymbol{B} \cdot \boldsymbol{E}$ is a compensated compactness quantity and therefore Relax($\mathcal{K}$) lies within the manifold $\boldsymbol{B} \cdot \boldsymbol{E} = 0$ (and hence has an empty interior). On the other hand, it can be shown that $\mathcal{K} \neq \text{Relax}(\mathcal{K})$. (A similar situation occurs in 2D MHD, see Faraco *et al.* 2021a, as well as stationary IPM, see Hitruhin and Lindberg 2021.) Initial investigations seemed to indicate that the empty interior was a serious obstruction, physically motivated as a manifestation of the conservation of magnetic helicity in this context (see Faraco and Lindberg 2018, Buckmaster and Vicol 2021). However, Faraco *et al.* (2021a) introduced a new convex integration scheme acting directly in two forms which is compatible with magnetic helicity conservation. Let us refrain from the subtle details but just indicate that the form formalism allows to identify those directions of the wave cone $\Lambda_g \subset \Lambda$ for which we could build Clebsch potentials compatible with the constraint $\boldsymbol{B} \cdot \boldsymbol{E} = 0$. A crucial geometric information is that in the 2-form formulation, $(\boldsymbol{u}, \boldsymbol{B}, \boldsymbol{E}) \in \Lambda$ implies that the corresponding form ω satisfies that $\omega \wedge \omega = 0$. But by a direct proof, the Plücker identities, this is equivalent to $\omega = a \wedge b$ for suitable a, b, that is ω is a simple form. For a simple and closed form, the correct Clebsch potentials can be found, except for some corner choices of a, b in relation to the direction of oscillation which has to be excluded. The proof is followed by a perturbation of the classical convex integration scheme adapted to Λ_g.

Elaborating on this idea, Faraco *et al.* (2021b) showed that indeed every piecewise constant solution to the Faraday–Maxwell system satisfying $\boldsymbol{B} \cdot \boldsymbol{E} = 0$ could be lifted to solutions to the full MHD system. More formally,

Theorem 4.1 (Lifting theorem): *Given a solution* $(\bar{\boldsymbol{E}}, \bar{\boldsymbol{B}})$ *to the Faraday–Maxwell system with* $\bar{\boldsymbol{B}} \cdot \bar{\boldsymbol{E}} = 0$ *there exist solutions* $(\boldsymbol{B}, \boldsymbol{u})$ *to the MHD system with "arbitrary" prescribed energy and cross helicity but* $\mathcal{H}(\bar{\boldsymbol{B}})(t) = 0 = \mathcal{H}(\boldsymbol{B})$.

The word "arbitrary" should be interpreted as "any two functions which satisfy the trivial definitions of energy and cross helicity densities".

With the lifting theorem at hand, by choosing carefully the solution to the Faraday system we could prescribe initial magnetic helicity which remains conserved, and prescribed energy and cross helicities which were dissipating, in accordance with what is seen in simulations. This is the idea behind the proof of Theorem 3.3.

The construction above yields bounded solutions. By Theorem 3.1 we know that magnetic helicity is preserved in the ideal limit and that L^3 solutions conserved magnetic helicity. As discussed in the intro, simultaneously with our investigations of bounded solutions, Beekie *et al.* (2020) discovered a discrepancy between solutions in the ideal limit and solutions of the ideal equations by showing that there exist $L^{2+\epsilon}$-integrable solutions to MHD violating magnetic helicity conservation. The beautiful paper uses the

theory of intermittency, similar to Buckmaster and Vicol Buckmaster and Vicol (2019a) on Navier–Stokes.

Following the philosophy explained above, we notice that $\boldsymbol{B} \cdot \boldsymbol{E}$ is no longer a compensated compactness quantity when $\boldsymbol{B} \cdot \boldsymbol{E} \notin L^1$ and hence the whole situation is more flexible.

Faraco *et al.* (2021b) looked at the relaxation of the Faraday–Maxwell system below the integrability threshold. The issue is to find solutions with integrability close to L^3. It is pertinent to this discussion to recall again that the theory of weak solutions to partial differential equations is tailored with a natural integrability exponent. The prime example is what is called Weyl's lemma for harmonic functions says that distributional harmonic functions whose derivative is in L^1 are indeed harmonic; the claim fails if less integrability is assumed. Indeed, the theory of convex integration in relation to integrability issues was developed by Astala *et al.* (2008) making strong use of the staircase laminates introduced by Faraco (2003) (see also Conti *et al.* 2005, Boros 2013). It has been used to obtain weak solutions below the natural integrability exponents, so-called *very weak solutions* (see Faraco 2004, Astala *et al.* 2008, Oliva 2016, Faraco *et al.* 2018). Elaborating on such ideas (e.g. we invented a curvy anisotropic staircase laminate) we obtain the following sharp result for the Faraday–Maxwell system.

Theorem 4.2: *Let $\bar{\boldsymbol{B}}, \bar{\boldsymbol{E}} \in L^\infty(\mathbb{T}^3 \times [0, T])$ be a pair of piecewise constant vector fields solving the Faraday–Maxwell system in the sense of distributions. Then there exist piecewise constant vector fields $\boldsymbol{B}, \boldsymbol{E} \in L^1(\mathbb{T}^3 \times [0, T])$ solving the Faraday–Maxwell system with*

$$\boldsymbol{B} \in L^\infty(0, T; L^{3,\infty}(\mathbb{T}^3)), \quad \boldsymbol{E} \in L^\infty(0, T; L^{3/2,\infty}(\mathbb{T}^3))$$

such that $\boldsymbol{B} \cdot \boldsymbol{E} = 0$ for a.e. (x, t) and

$$\mathcal{H}(B)(t) = \mathcal{H}(\overline{B})(t) \quad \textit{for a.e. } t.$$

Theorem 3.2 follows by combining Theorems 4.1, 4.2, and thus the sharpness of magnetic helicity conservation holds. It is intriguing that for weak solutions, the physically natural phenomenon "$\boldsymbol{B} \cdot \boldsymbol{E} = 0 \implies \partial_t \mathcal{H}(\boldsymbol{B}) = 0$" breaks down precisely when one exits the natural integrability regime $\boldsymbol{B} \in L^3, \boldsymbol{E} \in L^{3/2}$ (or when $\boldsymbol{B} \cdot \boldsymbol{E}$ ceases to be a compensated compactness quantity).

For other works dealing with other aspects of MHD, e.g. compressible, Hall, viscous and hyperviscous MHD as well as MHD forced by random noise, via convex integration see Dai and Liu (2019), Feireisl and Li (2020), Klingenberger and Markfelder (2020), Dai (2021), Dai and Friedlander (2021), Yamazaki (2021) and Li *et al.* (2021).

4.3. Macroscopic variables in ideal MHD and magnetic reconnection

We conclude with some comments on the contribution of the above concepts in understanding the scenarios allowing magnetic reconnection. We emphasise once more that apart from the existence of unexpected turbulent solutions, another appealing aspect of convex integration (or more precisely the Tartar framework) is that it provides additional information on the macroscopic behaviour of turbulent flows. As explained above, we can take weak limits, they satisfy the conservation laws but break down the constitutive

laws. Nevertheless, Tartar's compensated compactness theory is able to identify nonlinear weakly continuous quantities. In this regard, recall that it has conventionally been agreed that non-ideality of the plasma flow, and in particular the condition $\boldsymbol{B} \cdot \boldsymbol{E} \not\equiv 0$, is necessary for magnetic reconnection to be possible (see, e.g. Priest and Démoulin 1995, p. 452). This view has been recently challenged (Eyink 2015). Eyink proposed, instead, that by writing $\boldsymbol{E} + \boldsymbol{u} \times \boldsymbol{B} = \boldsymbol{R}$, the term $\boldsymbol{R}$ might be very small but instead, for magnetic reconnection to happen, $(\nabla \times \boldsymbol{R}) \cdot \boldsymbol{B}$ should fail to be small. The solutions of the papers by Faraco *et al.* (2021a, 2021b) are consistent with this latter hypothesis of Eyink, although as pointed out above, they deal directly with the ideal equations.

5. Discussion

We have presented a number of results on the behaviour of turbulent solutions to MHD via the concept of weak solutions. On the positive side, we presented a mathematical proof of Taylor conjecture for simple and multiple connected domains. We then describe the existence of bounded solutions which dissipate energy and cross helicity. Such solutions necessarily must preserve magnetic helicity but it turns out that as soon as the solutions are allowed to lie below the integrability threshold there are examples where magnetic helicity preservation is no longer true.

The above investigation leads to a number of interesting open problems, from the point of view of physics and mathematics, that we quickly list.

Firstly, we did not attempt to improve the differentiability of the solutions. It is natural to ask whether the regularity of energy dissipative weak solutions can be pushed to the $C^{1/3}$ threshold (Buckmaster and Vicol 2021, Conjecture 2.10). Secondly, the result of magnetic helicity conservation already reveals that the ideal limit and solutions to the ideal equation might behave differently. At the moment we do not know about energy dissipation in the ideal limit.

In this article, we focused on the three-dimensional situation. The 2D case is special and not fully understood. On one hand, we have proved that mean-square magnetic potential is at least as robust a conserved quantity as magnetic helicity. Simulations seem to indicate that there should be two-dimensional turbulent solutions dissipating energy but so far we have been unable to attend it. Whereas in 3D, the nonlinear constraint $\boldsymbol{B} \cdot \boldsymbol{E} = 0$ of the relaxation still leaves some room for error terms in a convex integration scheme, in 2D, ideal Ohm's law itself extends to the relaxation, making convex integration highly challenging.

It is also an intriguing problem to prove analogues of Theorem 3.3 for other models encompassed by the self-organisation conjecture. The conservation part at the inviscid limit has been rigorously confirmed for energy in 2D Euler, Hamiltonian in SQG and mean-square magnetic potential in 2D MHD. To our knowledge, however, only Faraco *et al.* (2021a) have run convex integration in an integrability regime consistent with the self-organisation conjecture, where some of the classical invariants are conserved and others are dissipated.

Finally, we mention that concerning conserved quantities a very interesting theme in which variants of magnetic helicity are preserved and the various versions of Taylor conjecture. Particularly interesting is the understanding of winding helicity and field line helicity (see, e.g. Prior and MacTaggart 2020).

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